Chapter 6

Contactless Monitoring in Newborn using Infrared Thermography: A review

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ABSTRACT

In neonatal care, the monitoring of temperature, heart rate, and breathing rate plays a pivotal role in determining the neonate's health status and medical diagnosis. However, the existing contact-based methods confront challenges such as discomfort, skin infections, and altering sleep patterns. Infrared imaging is very useful in temperature related physiological changes measurement with the characteristic of non-contact, non-radiative and functional method.

This review paper describes and explains infrared thermography used in the monitoring of vital physiological parameters of neonatal. Due to its use as portable and reliable, the infrared thermography will be easily implemented for monitoring the neonate's body surface temperature and breathing rate. The review study indicates infrared thermography as a new era of research in neonate monitoring and has a broad future in medical imaging.

KEYWORDS: Infrared Thermography, Neonatal Monitoring, Medical Diagnosis.

6.1. INTRODUCTION

The first four weeks of life is crucial time for the newborn, termed as neonatal period, needs extra care [1]. Globally, neonatal mortality is declining during the last four decades. However, the neonatal mortality rate is declining slowly in most low-income countries, including India, and progress towards achieving Millennium Development Goal 4 with the objective of reducing child mortality [2], is on a moderate scale. To speed up progress, it is important that preventing neonatal deaths has been put in order of importance. Neonatal monitoring is the process of monitoring key physiological changes in infants (newborns) born with various conditions, including premature birth, infection, surgical complications, breathing and heart anomalies. Newborns who face complications due to prematurity or infection may have long-term problems and are also required to be monitored for a long period of time after delivery.

The newborn regulates its own body temperature much less efficiently than adults and therefore easily cools down or heat up much faster—the smaller and more premature the baby, the greater the risk. Keeping up a stable-micro environment is one of the key physiologic difficulties that an infant must face after delivery. Thermal stress in a newborn has been identified as a key cause of increased morbidity and mortality, emphasising the importance of early diagnosis by thermoregulation monitoring [3]. The World Health Organization (WHO) has issued a practical guideline for newborn thermal safety. It suggests a variety of linked techniques to promote neonatal thermal preservation, which is called as 'warm chain' [4]. It is also required that the monitoring system is not only limited to neonatal intensive care unit (NICU) but also portable and suitable for keep monitoring in different interlinked procedures of 'warm chain'.

Respiratory distress is one of the most common cause for a newborn to be admitted to the neonatal intensive care unit (NICU) and need continuous monitoring of breath rate and respiratory behaviors [5]. However, respiratory distress monitoring is not limited to premature newborns. It is increasingly common in postneonatal and modest prematurity.

For neonatal monitoring, many contact-based technologies have been suggested [6-10]. Raj et al. [11] suggested a simple, cost-effective technology but encountered an issue with the sensor's adhesion. Contact-based methods introduce a number of complications, including pain, the possibility of skin injury, and infections [12]. Additionally, for premature babies or newborns, there is the risk of skin damage from the detachment of the electrode. It emphasizes the importance of implementing a contactless approach that could be used in clinical settings. Contactless monitoring has attracted increased interest in newborns and appears to be a promising choice for addressing the difficulties associated with traditional contact monitoring [13]. In the non-contact method, the body temperature and breathing rate are measured remotely, without any physical contact with the body. Contactless breathing monitoring systems have evident advantages [14, 15]. It improves the patient's comfort (particularly for long-term monitoring of neonates and rehabilitation patients) and reduces the possibility of inaccurate measurement as distress caused by the contact device or adhesive sensor may alter the breathing rate and body temperature measurement. One of the non-contact techniques for breathing rate is Doppler radar [16]. This approach is usually subjected by the radiation, the bulkiness, and the extreme susceptibility to motion artefacts. Recently, the focus is broadened by introducing imaging techniques: visible [17], remote-photoplethysmography (PPG) [18], and thermal imaging [14].

Every object having a temperature above absolute zero temperature emits electromagnetic waves in the infrared range called thermal radiation. Infrared (IR) thermography, simply called thermal imaging, records the infrared radiation emitted by the body and, by using this radiation profile, visualises the temperature profile of the body. The thermal camera captures this thermal profile either in white-hot or in different color palettes. Thermal imaging can easily illustrate the variation

in body surface temperature, making it useful for the early identification and diagnosis of several diseases. In Medical diagnosis, thermal imaging is a non-contact, non-invasive, non-radiative method for analyzing the physiological function related to the thermoregulation of the body or organ temperature. With the advancement of infrared sensors, optics, electronics, and software, it has been used in the detection and assessment of various diseases like breast cancer, arthritis, diabetic, wound, inflamed joints. However, there are still many challenges of thermal imaging in medical diagnostic research and in developing a controlled environment for accurate acquisition [19]. Additionally, integration with other technology is also a challenge for researchers.

In this chapter, the scope of contactless body surface temperature and respiration monitoring using infrared thermography (IRT) in the newborn is reviewed. Section 6.2 describes the working of IRT, the history and evolution of the IRT cameras. Section 6.3 reviews the use of IRT in medical diagnosis. Section 6.4 reviews the IRT in newborns. Finally, the scope of this work and future direction is explained in Section 6.5.

6.2. INFRARED THERMOGRAPHY

6.2.1 Historical Perspective

Traditionally, the human measured the body temperature by using thermistors, thermocouples, and thermopiles put on the specific area of the skin. However, these sensors are difficult to attach and are slow to respond. Additionally, it is not possible to simultaneously measure on the adjacent skin surfaces. Also, the sensor contact is susceptible to any form of movement. In contrast, a thermal camera can remotely measure the entire body surface temperature, which is visualized in the two-dimensional plane.

Sir Herschel [20] was the first to discover the wave range beyond the visual range that can be used to quantify temperature or heat profiles. Sir John Herschel, the son of William Herschel, delivered the main temperature or heat illustration in 1840 from sun-oriented radiation [21]. He defined a term for it called 'thermogram,' which is also used today to illustrate the temperature distribution in an object. In the earlier days, thermograms were not visually pleasing, took a long time to respond, and were only available in black and white. The evolution of thermal cameras has a major role in the improvement of quality in thermal imaging. Earlier research in infrared thermography was mainly focused on the development of sensors with aiming efficiency as a major factor. The application of the thermal imaging was also limited to military and defence purposes. The technology was classified and highly secret during that time period. Lawson documented the first thermal image processing for a cancer patient by observing the temperature difference of cancer spots from other normal parts [22]. Nowadays, researchers and physicians are using thermography to explore a variety of different illness problems, including breast cancer, skin cancer, and thyroid disease, etc.

6.2.2 How IRT works

A thermogram or thermal image depicts how much infrared radiation an object radiates, transfers, and reflects. Thermal images are used to depict temperature variation because the amount of infrared radiation emitted increases as the temperature rises. Infrared radiation, which is found between visible and microwave, is part of the electromagnetic spectrum. The electromagnetic spectrum is partitioned into different wavelength groups, termed as bands, that are recognized by the strategies utilized to create and identify radiation inside each band.

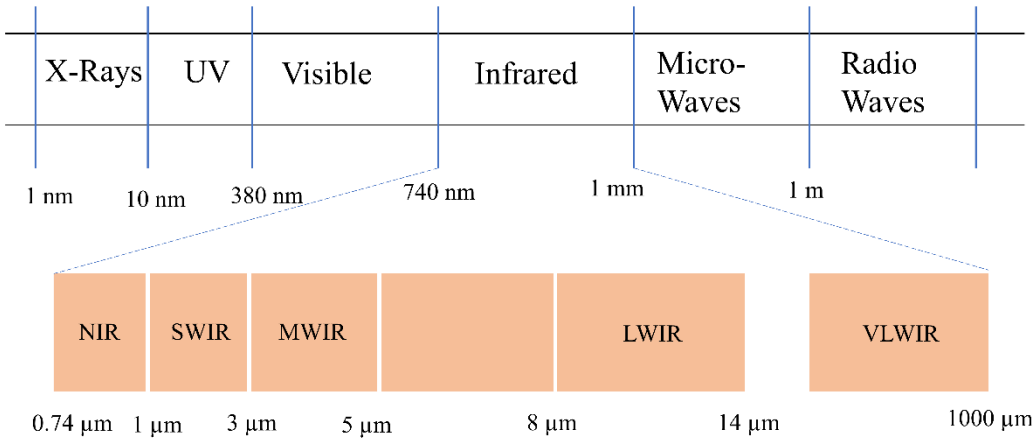

FIGURE 6.1 The visualization of a specified subdivision of infrared range in the electromagnetic (EM) spectrum.

These groups are called X-ray (<10nm), ultraviolet (UV) (<380nm), visible (<780nm), infrared (<1mm), microwave (<30cm), and radio-wave (<10 km, 30 kHz) bands (Figure 6.1). Further, the infrared is classified as near-infrared (740 nm–1 µm), short-wave infrared (1-3 µm), mid-wave infrared (3–5 µm), and long-wave infrared (8–12 µm). Thermal imaging measures the infrared radiation emitted by a body in the 8–14 µm (LWIR) range and illustrates the temperature profile of that body's surface [23]. The thermal camera or sensor is used to record the thermal profile of the object in the form of an image. The image represents the quantified temperature or radiation value of emitted infrared radiation at each spatial position of an object. The radiation energy captured by the thermal camera to determine the temperature is dependent on the body's temperature and emissivity. The ratio of an object's emitted energy to the emitted energy of the black body at an equivalent temperature is defined as the emissivity.

The black body can radiate maximum energy and work as the reference to specify the body. In the wavelength range of 2 to 14 micrometres, the emissivity of human skin is about 0.97–0.98 [24]. In general, the infrared camera has an optical system, one or more sensor mechanisms and detectors to convert captured infrared energy into electrical energy, some systems for scanning mechanism (currently, there is no requirement of this), a processor, and a display unit as shown in Figure 6.2. The working of an optical system is the same as a visible camera optical system, however, the

material is specific for infrared wavelength. The sensor mechanism (thermal detector) is the key element of the thermal camera. Nowadays, the photon or quantum detectors are usually referred, and for details about the technology, one can refer [25]. Some pre-processing is performed by the processor inbuilt in camera to set the thresholding, nonuniformity correction, pseudocolouring, etc. Typically, the image captured by the thermal camera is viewed on a separate external monitor, which could be a computer screen or a small screen embedded with camera.

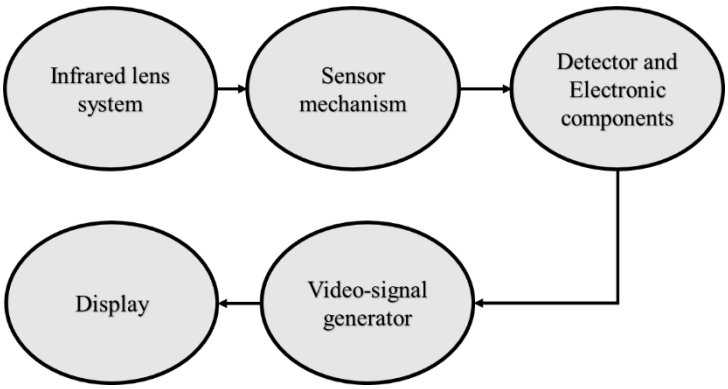

FIGURE 6.2 Overview of the components in thermal camera [26].

6.2.3 Evolution of Infrared Cameras

The common purpose of all thermal cameras is to transform infrared radiation into visual images [27]. The earliest generation of thermal cameras were big in size and included single element detector. This was improved by utilizing the array of around 120 elements. Later, a camera having detector elements arranged in rows and columns was used to obtain the two-dimensional visualisation. The latest generation cameras upgraded the accuracy and performance by utilizing large size focal plane array (FPA) and integrating of image computing capabilities [28, 29]. These have predominant temperature affectability, speed of operation, and spatial determination than prior adaptations [27]. The first-generation thermal cameras were huge, cooled with liquid nitrogen, requiring liquid nitrogen cooling, and had to be kept horizontal to prevent nitrogen from falling on the subject [23]. Currently, available cameras are smaller, simple to manage, free from orientation, and there is no need of liquid nitrogen because no refrigeration is required. Thermal cameras can produce both videos and digital images of any object. The video records the temperature variation of any object with time.

6.3. INFRARED THERMOGRAPHY IN MEDICAL DIAGNOSIS

Despite the fact that infrared thermography has been used by the researchers in the aviation, military, and mechanical industries, it is relatively new in the medical field. The thermal image of the human body illustrates the body's temperature anomaly. Medical diagnosis is based on structural and

functional information of the patient, as well as subjective symptoms. In general, structural information is obtained through imaging techniques such as ultrasound, classical microscopy, X-ray, and magnetic resonance imaging (MRI). Functional methods are like an electroencephalogram (EEG), pulmonary air flow tests (PFT), thermal imaging, etc. Infrared thermography is a contactless, non-invasive, and non-radiative technique for measuring the physiological function related to the body's thermal regulation or organ temperature in medical diagnostics.

Clinicians and researchers have recently studied novel and interesting applications with a wide range of clinical consequences, including the study of asymmetric body temperature distributions, which may play a role in the pathophysiology of a variety of diseases [30-33]. Additionally, the distribution of body temperature contributes as a measure of emotions and expression [34, 35]. Despite the limited investigation of thermal imaging in the paediatric populace particularly on neonates, numerous of the thermal imaging-based methodology utilized in grown-ups have potential and may be well suited for newborn children.

The human body's core behaves as a thermal reservoir, and a reasonable temperature remains under normal circumstances. Heat transfers from the bloodstream to the skin through the blood vessels. The rate of heat exchange between the core and the surrounding environment via radiation, conduction, convection, and evaporation regulates the temperature of the skin's surface [36, 37]. A normal healthy adult should have symmetrical skin temperature distribution across the body. Thermal Infrared thermography may reveal the asymmetry in temperature distribution caused by the perfusion abnormalities. [38-40]. Moreover, it is useful in the study of medical conditions related to changes in body temperature induced by vasodilation or constriction, hypothermia or hyperthermia, hypoperfusion or hyperperfusion, hypermetabolism or hypervascularization [38, 39].

In specific, researchers and doctors have used infrared thermography to detect and evaluate cancerous tumors. Mital et al. [33] developed a techniques that uses infrared thermography to assess the depth and location of a breast tumour. Another application of thermal imaging is the detection of fever when mass screening occurs during pandemic like situations. Clinicians utilized this technology in China during the peak of pandemic for the purpose of mass screening [41, 42]. Since thermal cameras are compact, light, and precise, it can rapidly measure the temperature of a large number of people without being exposed to people who are infected by potential infectious diseases.

Researchers have too utilized infrared imaging to analyze thyroid infections [38], affirm hydrocephalus shunt patency [43], evaluate the treatment of diabetic wounds [31], and analyze and diagnose leg ulcers [32]. In sport studies, analysts were utilized infrared thermography to map out temperature disparities during work out and rest [44].

6.4. INFRARED THERMOGRAPHY IN NEONATES

R. B. Barnes [45] investigated thermal imaging as a new concept for contactless temperature monitoring of the human body using infrared-radiant energy. His research hypothesised that by recording infrared radiation as a thermogram, he might build a precise temperature map of the skin. The skin temperatures were mostly derived by the vascularity of the skin and the heat transferred from within the body if the naked patient kept stationary in a cool environment for ten to fifteen minutes prior to thermal imaging. The resulting temperature map provide information on specific pathological states within the body, because the contrasts that develop under these settings are primarily due to internal sources of heat.

Clark and Stothers [46] pioneered the use of infrared thermography on neonates in the year 1980. They used infrared thermography to determine the distribution of neonatal skin temperature. They found that the arms, thighs, and buttocks of newborn have a mean value and are susceptible to changes in ambient temperature. They concluded that the skin covers the upper abdomen is heated and generally insensitive to changes in ambient temperature than the mean skin temperature.

Following that, more researchers focused on the scope of thermal imaging on newborns. Anderson et al. [47] investigated the use of thermal imaging to study children who sleep at home. First, the body temperature was recorded, and thermography was performed on five children, four of whom were sleeping in similar conditions for ambient temperature and isolation, and resultant thermography patterns was similar to deep body temperature. The other slept in more hotter conditions and made a diverse thermogenic reaction [48]. They concluded the potential possibilities of thermal imaging to observe the body temperature variations and body movements at home.

Adams et al. [49] developed a novel method for demonstrating the feasibility of using infrared thermographic calorimetry (ITC) to assess the energy expenditure of premature babies, as compared to the respiratory indirect calorimetry. ITC is a measurement of the average surface temperature of the child's body. The ITC has been used in conjunction with the heat loss theory to compute radiant heat losses, convection, evaporation, and conduction losses, thus determining the total energy consumption. They concluded that the ITC was an accurate and non-invasive approach for determining heat loss and energy expenditure in healthy premature infants and hence may be used as a clinical and diagnostic tool.

Furthermore, Christidis et al. [50], conducted an interesting study on infrared thermography in neonates during the first hour after childbirth. They measured the surface temperature of a newborn from the time of birth to the first hour. Additionally, the impact of a variety of environmental factors on surface temperature has been demonstrated. Thermal imaging is a simple tool for depicting a person's temperature profile, according to their research. It could be useful to optimize the environmental conditions in delivery rooms and newborn critical care units, without requiring direct

contact with the skin. However, thermal imaging revealed the newborn's whole body surface temperature but not the core temperature.

Further, the infrared thermography was used to measure the skin temperature of the newborn on different positions and postures, including kangaroo maternal care [51]. IRT is also a valuable strategy for non-contact physiological monitoring of breathing. By using IRT, it is possible to record the average thermal airflow of the patient's nostrils, related to breathing. The nostrils are recognised and defined as a region of interest (ROI), which is then automatically tracked throughout the measurement procedure employing image processing methods. This technology is a promising option for passive contactless monitoring of respiratory activity in preterm infants and may be used in conjunction with infrared thermography-based newborn temperature monitoring.

In recent years, the researchers investigated the potential, advances and limitation of thermal imaging for the purpose of breathing monitoring. A study on premature babies for respiration monitoring using infrared thermography measurement has been reported by Abbas et al. [52]. They manually selected the nostril region as ROI for respiration rate detection. A pilot study was performed in [53], in which a simple MATLAB code was used to calculate respiration rate (RR) by using infrared imaging. The study's validity was established by the use of an ECG-based technique. This had an excellent agreement in visual results [53]. Another noteworthy technique performed on premature neonates [54] proved reliable. It tracked newborn motion using a particle-filter-based object tracker. A Butterworth filtering was performed for further processing and RR evaluation. The methodology is so easy and feasible and it has a high-level agreement. Likewise, Pereira et al. [5] proposed a technique for newborn infants that used the "Black Box Algorithm" for automatic ROI detection. Since there is no longer anatomical structure to assist in facial recognition, this algorithm works even without a frontal view as compared to the work reported in [52].

Researchers at Carleton University in Canada performed a study on thermal imaging for the diagnosis of Necrotizing Enterocolitis [55]. This study was demonstrated a novel method of abdominal infrared thermal imaging that capable of intestinal inflammation characteristics of Necrotizing enterocolitis (NEC). The result hopefully suggested that this new non-contact, non-ionizing method could be used as an early diagnostic tool [55].

6.5. CONCLUSION AND FUTURE DIRECTIONS

Infrared thermography is an innovative method for studying body surface temperature and tissue perfusion in adults, children, and newborns in both research and clinical settings. This technique is non-invasive and convenient (due to advancement in sensor technology) and does not expose neonates to hazardous radiation. The potential for new research with infrared imaging is growing rapidly. For example, nursing researchers can use thermography to monitor wounds and identify

skin temperature variations associated with pressure injuries, as the temperature differential between normal skin and skin that is deteriorating due to infection can be detected long before visual examination reveals discoloration as a result of impaired skin integrity.

Neonatal researchers can apply this technology in a variety of areas, including studying kangaroo care, which involves the transfer of heat between the mother's breast and the wrapped baby, cooling the body for hypoxic-ischemic encephalopathy treatment, and NEC.

In NICU, body temperature, electrocardiogram (ECG), respiratory rate, and blood oxygen saturation are the primary vital indicators of clinical significance for continuous neonatal monitoring. Presently, in the neonates, the body temperature is measured by the adhesive thermistor and also ECG, respiration rate are measured by adhesive skin electrodes. The placement and removal of these adhesive skin sensors (electrodes) and the presence of all wire are discomforts and even painful stimuli. Accordingly, the most important cause of skin breakdown is removal of sticky electrodes from newborns in intensive care units. Hence, an alternative, non-invasive, non-contact method is needed for the monitoring of vital physiological function to provide neonatal care and monitoring. The thermal imaging has demonstrated the potential for accurate temperature and respiratory detection. However, there are still many challenges and concrete pieces of evidence needed in terms of clinical evaluation.

Infrared thermography-based innovation has the potential to reduce neonatal therapeutic costs and length of stay for newborns who have illness like NEC or hypoperfusion, or who have surgery to fix a problem with their stomachs. This innovation is in its earliest stages, and chances for research abound. As researchers continue to study this prospective technique, they will uncover a variety of other research and therapeutic applications.

In the future, thermal image-based monitoring is not limited to neonatal monitoring but is also used for high-risk patients during the critical operating phase. For that, a lot of research work, clinical trials, and case studies are needed to make it feasible for life risk. Thermal imaging represents a new era in medical research with a bright future.

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